

Wave 12 — CJ Inventory Intelligence

Delivery & production deployment report · trovemo.com · 2026-07-28 · branch `fix/goods-management` → `master 953ded741`

1. Executive summary

Wave 12 gives the store true cost visibility and margin control over its CJ Dropshipping catalog. Until now the raw CJ wholesale price was **discarded at sync**: products were priced with a legacy `× 7.2 currency × 2.0 margin` formula (a flat `×14.4`), leaving an average \$260.91 retail on roughly \$18 cost and no recorded margin anywhere. Wave 12 persists the wholesale cost end-to-end, reprices the store at **retail = cost × 1.25** (user decision 2026-07-28 — a deliberate ~91% storewide price reduction), tracks every product daily from its arrival date, scores daily new arrivals into admin-approved deal proposals, and re-opens flash deals for CJ goods with a hard floor at real cost. All backend flows follow Spring Integration patterns (Fisher et al., *Spring Integration in Action*) behind the existing nightly sync.

2. What was delivered

Area	Delivery
V45 migration	<code>litemall_cj_product.sell_price</code> (raw CJ USD cost) + 3 new tables: <code>litemall_cj_sync_run</code> (per-phase pipeline bookkeeping), <code>litemall_product_metric_daily</code> (idempotent per-goods/per-day price·cost·margin·stock·availability·views·sales series), <code>litemall_deal_candidate</code> (proposed approved dismissed, unique per goods+day).
Cost capture & ×1.25 repricing	One shared <code>CjPricing</code> component replaces three duplicated formula sites; the <code>×7.2 currency</code> factor was deleted , not set to 1. Cost lands on the snapshot, the goods row and every SKU (per-variant <code>variant_sell_price</code>); the long-dead <code>V2 cost</code> columns are now mapped and written. Margin fields render <code>null</code> — never a fake 0 — until a product's cost is captured.
Inventory flow (Spring Integration)	Messaging gateway fed by the nightly catalog cycle and by every enrichment path; splitter → content enricher → router (<code>NEW_ARRIVAL</code> / <code>UPDATED</code> / <code>VANISHED</code>) → service activators (margin recorder, availability recorder, deal-candidate scorer) → per-category aggregator into a memoized rollup; wire tap writes <code>litemall_cj_sync_run</code> . A nightly 04:15 re-check queues deal-relevant variant ids into a bounded channel drained at 1 msg/s through <code>CJ.getInventory</code> — the global CJ 1 req/s budget is respected and no CJ call ever runs on a request path .
Admin insight API	<code>/srv/private/admin/insight/**</code> per the Wave-12 contract: category rollup ranked by potential profit; per-goods list with server-side sorting on arrival, price, margin, stock and sales; per-goods dossier (variants with per-SKU cost, 90-day series, engagement totals, honest "advertisable" recommendation); deal-candidate list + approve/dismiss (errno 653). Consumed by the gateway-admin Wave-12 insight UI.

CJ flash deals unparked	The blanket errno-652 refusal became a floor at the real captured cost (below-floor → 650 naming the floor; 652 narrowed to "cost unknown"). While a deal is live the CJ promote path withholds price writes; unwind re-promotes from the snapshot; the PDP deal endpoint resolves <code>cj_<pid></code> references. Deal proposals are only ever created by admin approval .
Charge-integrity guard (critical catch)	Found during dev e2e: the sync never rewrites enrichment-owned variant data, so after repricing the goods row displayed the new price while checkout would still have charged the old ×14.4 SKU price storewide . The adapter now discards old-basis variant prices when a captured cost exists, charging the product retail until enrichment recomputes true per-variant prices. Without this, the first production sync would have mispriced checkout across the entire catalog.

3. Verification before deployment

Gate	Result
Module test suite	102 run / 0 failures (8 skips = live-OCS probes by design)
Migration validation	FlywayMigrationTest 4/4 — all 45 migrations on a fresh MySQL container (test converted to JUnit 5; the module's JUnit-4 suite had been silently non-running)
Dev pricing e2e	sync ⇒ <code>sell_price</code> captured, price/cost ratio 1.2496-1.2505; PDP via :9000 = 14.81 = 11.85 × 1.25; SKU price == displayed price after guard (was 170.64)
Pipeline bookkeeping	sync/flow run rows recorded; incomplete sample fetch honestly marked <code>complete=0</code>; promote 9,644/9,644, reindex 9,644 docs
Insight API via :18080	categories ranked by potential profit; all 5 sort keys round-trip server-side; series + per-variant costs + recommendation
Deal lifecycle e2e	NEW_ARRIVAL scored featured (92.16) → admin approve ⇒ CJ deal live in ≤60s (SKU 6.64 ≥ cost 6.25, originals captured) → disable ⇒ byte-identical restore + snapshot re-promote
Checkout regression via :9000	cart → submit → wallet pay: order paid (status 201) at the repriced amounts; un-enriched CJ item correctly blocked by the typed retryable 422

4. Production deployment — trovemo.com

Step	Outcome
Merge	fix/goods-management → master fast-forward 308082607..953ded741 (2 commits, 63 files, +3,748/-152); pushed to GitHub
Ship to VPS	pushed to <code>/opt/litemall.git</code>; checkout fast-forwarded to 953ded741
Image build	litemall/goods-management:latest built on the VPS (50G free ≥ 30G build threshold)
Recreate	container recreated behind healthy eureka/config/mysql/redis/rabbitmq/elasticsearch dependencies

V45 on production DB	"Successfully applied 1 migration ... now at version v45" (0.176s); service started in 9.1s
Edge health	https://trovemo.com 200; /srv/goods/index serving real payloads after Eureka re-convergence
Storewide repricing (first live sync)	LIVE — two cycles completed (18:29 & 20:08 UTC): 3,498 of 11,682 products cost-captured so far; average repriced retail \$24.82 (previously ~\$260.91); 254 deal proposals generated for admin review

4.1 First live repricing cycles — measured results

Metric	Cycle 1 (18:12-18:29)	Cycle 2 (19:50-20:08)
Snapshot sync (new / updated / removed)	244 / 3,256 / 0	10 / 3,089 / 0
Native promote	11,672 promoted / 0 failed	11,682 promoted / 0 failed
Search reindex	11,672 docs	11,682 docs
Pipeline bookkeeping (litemall_cj_sync_run)	4 rows (sync + flow per cycle), all complete = 1	
Daily metric rows (litemall_product_metric_daily)	3,498 products tracked from day one	
Deal proposals (litemall_deal_candidate)	254 proposed — awaiting admin approve/dismiss (never auto-created)	
Formula integrity	3,490 of 3,498 costed goods exactly at cost × 1.25 (±1¢); 8 transitional rows — see below	
Erosion protection	reconcile tripwire correctly VETOED a rotation-limited delete of 8,613 goods (74% > 20% threshold) — zero products removed	

Live finding & same-day hotfix (commit 6dc10d36a). CJ's listing rotation returns only ~3,500 of the 11,682 products per cycle, so most of the catalog reprices over successive nightly syncs. Minutes after the first deploy, customer PDP views triggered on-demand enrichments that captured a product's cost *without* recomputing its retail — leaving a handful of products at the old ×14.4 price next to a fresh cost. Enrichment now reprices from the captured cost, making it a proper repricing path for the rotation tail. The 8 rows created during the pre-fix window are overpriced (never underpriced — no revenue risk) and self-converge via the nightly sync/enrichment; no new such rows can occur.

Catalog coverage outlook. Costs (and the ×1.25 price) now flow in through three converging paths: the nightly 03:00 sync (whatever CJ's rotation lists), the nightly 03:30 enrichment cursor (views-prioritized), and on-demand enrichment when customers view a product. Products without a captured cost keep their previous price and show `null` margins until their cost lands — never a fake number, and CJ deals on them stay refused until the cost basis is known.

5. Operational notes

- **Costs populate as syncs run.** Products not yet re-synced keep `cost = not captured`; their margins render as `null` in every surface and CJ deals on them are refused (652) until the cost lands.
- **Scheduler change:** an explicit 4-thread scheduler replaces the implicit single-thread one — the nightly CJ refresh no longer starves the 60-second deal-lifecycle tick.
- **Per-variant prices during transition:** products whose variants predate the repricing charge the product retail (uniform $\times 1.25$) until the nightly enrichment recomputes per-variant prices from per-variant costs.
- **Nightly cadence:** 02:45 related-goods · 03:00 catalog sync → flow → rollup · 03:30 enrichment · 04:15 deal-goods inventory re-check (1 req/s, bounded queue).
- **Sibling deliveries:** the gateway-admin Wave-12 insight UI consumes this API; the promotion worktree ships the category-campaign scheduler. Both code to the frozen Wave-12 contract.

Generated 2026-07-28 · Wave 12 goods-management worktree session · commits `8bc1e83a8` (backend core) + `953ded741` (charge-integrity guard) · production: Hetzner VPS, 19-container compose stack, Cloudflare-proxied edge.